Quantum Cosmology of Successive Universes

Aarav Shah*[1] and Kanabar Jay†[2]

[1]Virtual Institute of Astroparticle Physics, Paris 75018, France
[2]M.G. Science Institute, Gujarat University, Ahmedabad 380009, India

October 27, 2026

Abstract

We propose a generalized framework for quantum cosmology in which successive spacetime manifolds are coupled through a common crossover hypersurface by extending the gravitational path integral with a geometric matching functional. As an illustrative example, we consider Penrose's Conformal Cyclic Cosmology, where consecutive universes are related by a conformal transformation. The resulting matching modifies the minisuperspace Wheeler-DeWitt equation, yielding an effective quantum potential governed by the conformal factor. Solving Penrose's Phantom Field Equation, we find that the conformal factor rapidly grows with the scale factor, while the effective Wheeler-DeWitt potential is rapidly suppressed. We further investigate cosmological perturbations and find that the conformal crossover preserves a nearly scale-invariant primordial spectrum with $n_s \simeq 1$.

1 Introduction

The Wheeler-DeWitt equation [1, 2] provides the canonical description of quantum cosmology [1–16], where the quantum state of the Universe is represented by a wave function satisfying the Hamiltonian constraint. In its conventional formulation, the wave function is defined on a single spacetime manifold, and the corresponding gravitational path integral is constructed by summing over geometries compatible with prescribed boundary data [14, 15]. Within this framework, the quantum state is associated with a single cosmological spacetime, without considering the possible influence of preceding cosmological universes.

A natural extension of this framework is to ask whether the quantum state of a universe could retain information inherited from an earlier cosmological universe. If spacetime consists of a sequence of successive cosmological manifolds connected through an appropriate transition hypersurface, it is conceivable that the quantum state of one universe may not be entirely independent of its predecessor. Such a possibility would require a generalization of the conventional path integral, allowing quantum amplitudes to depend on data inherited from an earlier spacetime.

In this work, we propose such a framework by considering two spacetime manifolds, $\mathcal{M}_1$ and $\mathcal{M}_2$, representing successive cosmological histories. We introduce an additional geometric functional that couples the two manifolds through a common crossover surface, thereby extending the gravitational path integral to include correlations between successive universes. Upto this point, the resulting construction is intentionally general and does not depend on a particular realization of the crossover geometry.

As a specific example, we consider the conformal matching conditions of Penrose's Conformal Cyclic Cosmology [17–20], in which the asymptotic future of one universe is conformally identified with the Big Bang hypersurface of the next. Within our formalism, this corresponds to a particular choice of the matching functional relating the metrics of the two manifolds. We then investigate the resulting quantum cosmology in

*Corresponding author:- shahaarav103@zohomail.in
†kanabarjaymgsi@gmail.com

minisuperspace, deriving the modified Wheeler-DeWitt equation and studying the evolution of the effective potential inherited through conformal rescaling. We further extend the analysis to cosmological perturbations [11, 21–23], suggesting that the conformally invariant branch preserves the primordial power spectrum across the crossover surface. Since the preceding aeon approaches an asymptotically de Sitter phase, the resulting scalar spectrum remains nearly scale invariant, with $n_s \simeq 1$ [24], suggesting a preliminary mechanism by which successive universes inherit approximately scale-invariant primordial perturbations.

2 Quantum Cosmology of Successive Universes

Let $\mathcal{M}_1$ and $\mathcal{M}_2$ denote two spacetime manifolds representing successive cosmological histories. We wish to investigate the possibility that the quantum state of $\mathcal{M}_2$ is influenced by geometric information inherited from $\mathcal{M}_1$. In the Hartle-Hawking prescription [14], the wave function of the Universe is defined through a gravitational path integral over four-dimensional geometries that induce a prescribed three-metric h_{ij} on a final spatial hypersurface Σ

$$\Psi = \int_{\Sigma} \mathcal{D}g_{\mu\nu} e^{iS}, \tag{1}$$

where we set $\hbar = 1$.

We retain this boundary prescription and extend the path integral to incorporate effects from the manifold $\mathcal{M}_1$. Their geometries are coupled through an additional crossover action, allowing geometric information from the preceding universe to constrain the geometry of its successor. The resulting wave function is therefore expressed as

$$\Psi = \int_{\Sigma} \mathcal{D}g^{\mathcal{M}_2}_{\mu\nu} \mathcal{D}g^{\mathcal{M}_1}_{\mu\nu} \mathcal{D}\lambda e^{\pm iS_{\mathcal{M}_2}[g^{\mathcal{M}_2}_{\mu\nu},\phi]} e^{iJ[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}]}, \tag{2}$$

where the integration is taken over all four-geometries compatible with the prescribed boundary data on Σ. The functional

$$J[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}] = \lambda S_{\text{cross}}[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}], \tag{3}$$

may be interpreted as an effective source term originating from the preceding spacetime $\mathcal{M}_1$, which couples the geometries of $\mathcal{M}_1$ and $\mathcal{M}_2$ and encodes the influence of the former on the quantum state of the latter. Here λ is an auxiliary Lagrange multiplier field, while S_{cross} is a geometric functional

$$S_{\text{cross}} = \int_{\Sigma} d^3x F[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}], \tag{4}$$

that characterizes the correlation between the two spacetimes. Integration over λ then yields,

$$\Psi = \int_{\Sigma} \mathcal{D}g^{\mathcal{M}_2}_{\mu\nu} \mathcal{D}g^{\mathcal{M}_1}_{\mu\nu} \delta(S_{\text{cross}}[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}]) e^{\pm iS_{\mathcal{M}_2}[g^{\mathcal{M}_2}_{\mu\nu},\phi]}. \tag{5}$$

Consequently, only those pairs of geometries satisfying

$$S_{\text{cross}}[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}] = 0, \tag{6}$$

contribute to the quantum amplitude.

Since the two manifolds communicate only through the crossover surface $\mathcal{C} = \mathcal{M}_1 \cap \mathcal{M}_2$, the matching condition must only be imposed on the metric restricted to $\mathcal{C}$. The simplest choice of this constraint is then

$$F[g^{\mathcal{M}_1}_{\mu\nu}, g^{\mathcal{M}_2}_{\mu\nu}]\big|_{\mathcal{C}} = g^{\mathcal{M}_1}_{\mu\nu}\big|_{\mathcal{C}} - \Omega^4 g^{\mathcal{M}_2}_{\mu\nu}\big|_{\mathcal{C}} = 0, \tag{7}$$

corresponding to Conformal Cyclic Cosmology [17–20].

The conformal matching condition immediately implies a corresponding rescaling of the FLRW scale factor across the crossover surface,

$$a^{\mathcal{M}_1}\Big|_{\mathcal{C}} = \Omega^2 a^{\mathcal{M}_2}\Big|_{\mathcal{C}}. \tag{8}$$

The quantum dynamics in the manifold $\mathcal{M}_1$ are described by the minisuperspace Wheeler-DeWitt equation,

$$H_{\mathcal{M}_1}\Psi(a) = \left[\frac{1}{24\pi^2}\frac{\partial^2}{\partial a^2} - 6\pi^2 V(a)\right]\Psi(a) = 0, \tag{9}$$

where the corresponding Wheeler-DeWitt potential is

$$V(a) = a^2 - H^2 a^4. \tag{10}$$

Using the conformal relation

$$a^{\mathcal{M}_2} = \Omega^{-2} a^{\mathcal{M}_1}, \tag{11}$$

the Wheeler-DeWitt equation may be rewritten in terms of the scale factor of the manifold $\mathcal{M}_2$. The Hamiltonian constraint therefore becomes

$$H_{\mathcal{M}_2}\Psi = \left[\frac{1}{24\pi^2}\frac{\partial^2}{\partial(\Omega^{-2}a)^2} - 6\pi^2 V(\Omega^{-2}a)\right]\Psi = 0, \tag{12}$$

where the transformed Wheeler-DeWitt potential is

$$V(\Omega^{-2}a) = \Omega^{-4}a^2 - \left(H - \frac{2\dot{\Omega}}{\Omega}\right)^2 \Omega^{-8}a^4. \tag{13}$$

Now, using the chain rule, we can write derivatives with respect to $\Omega^{-2}a$ in terms of derivatives with respect to a,

$$\frac{\partial}{\partial(\Omega^{-2}a)} = \frac{\Omega^2}{(1-2Fa)}\frac{\partial}{\partial a}, \tag{14}$$

where

$$F = \frac{\dot{\Omega}}{aH\Omega}. \tag{15}$$

Similarly, we have

$$\begin{aligned}\frac{\partial^2}{\partial(\Omega^{-2}a)^2} &= \frac{\Omega^4}{(1-2Fa)^2}\frac{\partial^2}{\partial a^2} + \frac{\Omega^2}{(1-2Fa)^2}\frac{\partial\left(\ln\left(\Omega^{-2}(1-2aF)\right)\right)}{\partial a} \\ &= J^{-2}\left(\frac{\partial^2}{\partial a^2} - L\frac{\partial}{\partial a}\right),\end{aligned} \tag{16}$$

where

$$J = \Omega^{-2}(1-2aF), \quad L = \frac{d\ln J}{da} = -2F - \frac{2\left(F + \frac{\dot{F}}{H}\right)}{1-2aF}. \tag{17}$$

The Wheeler-DeWitt equation may be recast into the equivalent form

$$\left[\frac{1}{24\pi^2}\left\{\frac{\partial^2}{\partial a^2} + \frac{1}{2}\frac{\partial L}{\partial a} - \frac{L^2}{4}\right\} - 6\pi^2 J^2 V(\Omega^{-2}a)\right]\Phi = 0, \tag{18}$$

where we have introduced the rescaled wavefunction

$$\Phi = J^{-1/2}\Psi = \frac{\Omega}{\sqrt{1-2aF}}\Psi. \tag{19}$$

Eqn.(18) can then be expressed in the standard form as,

$$\left[\frac{1}{24\pi^2}\frac{\partial^2}{\partial a^2} - U(a)\right]\Phi = 0, \tag{20}$$

where the modified Wheeler-DeWitt potential ($U(\Omega^{-2}a)$) is given by

$$U(\Omega^{-2}a) = \frac{1}{96\pi^2}\left[L^2 - 2\frac{\dot{L}}{\dot{a}}\right] + 6\pi^2\Omega^{-4}(1-2aF)^2\left[\Omega^{-4}a^2 - \left(H - \frac{2\dot{\Omega}}{\Omega}\right)^2\Omega^{-8}a^4\right], \tag{21}$$

Now, the conformal factor Ω is determined, by the Phantom field equation [18, 25]

$$\left(\Box + \frac{1}{6}\tilde{R}\right)\Omega = \frac{1}{6}R^{(\mathcal{M}_1)}\Omega^3, \tag{22}$$

where $R^{(\mathcal{M}_1)}$ denotes the Ricci scalar of the manifold $\mathcal{M}_1$, while $\tilde{R}$ is the Ricci scalar associated with the intermediate metric defined on the conformal crossover region.

Penrose proposes that this must be satisfied with the restriction $R^{(\mathcal{M}_1)} = 4\Lambda^{\mathcal{M}_1} = \tilde{R}$, where $\Lambda^{\mathcal{M}_1}$ is the cosmological constant. Newman [26], Tod [25], Nurowski [27] all differ from Penrose by not enforcing $R^{\tilde{n}} = R^n$, but rather that R is possibly some other constant.

For concreteness, we adopt Penrose's original prescription [18] and, for numerical purposes, set the scale factor when the conformal regime is reached to the representative value of $a_i = 10^4$. At this stage, the Universe has entered the asymptotic future of the manifold $\mathcal{M}_1$, where all massive degrees of freedom have decayed or effectively become massless, rendering the large-scale dynamics conformally invariant. Now, when the process of conformal rescaling starts at a_i, the conformal factor must start from unity so that $\Omega(a_i)^{-2}a_i = a_i$. However, it may have any generic value of $\frac{d\Omega(a_i)}{da}$, which motivates the initial conditions

$$\Omega(a_i) = 1, \quad \frac{d\Omega(a_i)}{da} = k. \tag{23}$$

The Phantom Field equation (22) governs the conformal factor only during the aeonic transition, in which the geometry of $\mathcal{M}_1$ is matched to the crossover surface $\mathcal{C}$. We introduce a critical scale factor a_c of the scale factor beyond which the transition is complete and $\mathcal{M}_2$ has "begun". For $a \geq a_c$, Ω is no longer dynamical and remains fixed at the value it has reached,

$$\Omega(a) = \Omega_c \equiv \Omega(a_c), \quad a \geq a_c. \tag{24}$$

Beyond the crossover surface $\mathcal{C}$, $\mathcal{M}_2$ evolves through its own Friedmann dynamics, and the crossover only provides initial data.

The numerical solutions of the Phantom Field Equation is shown in Fig.1, in the transition region $a_i \leq a \lesssim a_c$ with the initial slope fixed at the values $k = 10^{-5}, 10^{-4}, 10^{-3}\text{and}10^{-2}$. Two qualitative features are common to both cases, and they are the only ones that matter for the present study. The precise, detailed behavior of the modified Wheeler-DeWitt potential in the intermediate region, including its non-monotonic features, is not relevant to our conclusions and is not analyzed further here.

The first feature is the extreme growth of the conformal factor. In both cases Ω stays close to unity over most of the early evolution and then grows explosively, rising by several orders of magnitude over a very narrow interval of a. The second feature is the extreme suppression of the modified Wheeler-DeWitt potential. Over the same interval, the modified potential falls by many orders of magnitude and becomes negligible compared with its initial value.

The physical consequence of this behavior is particularly interesting from the perspective of quantum cosmology. As the second universe of $\mathcal{M}_2$ exists the crossover surface, the potential term appearing in the Wheeler-DeWitt equation becomes negligible compared with the kinetic term. Equivalently, the conformally redefined wave function, $\Phi = J^{-1/2}\Psi$ evolves as an approximately free Wheeler-DeWitt wave function while traversing into the second universe of $\mathcal{M}_2$. In this sense, the crossover surface acts as an asymptotically free quantum boundary between successive universes, where the quantum state propagates with negligible influence from an effective minisuperspace potential. We now investigate the behavior of cosmological

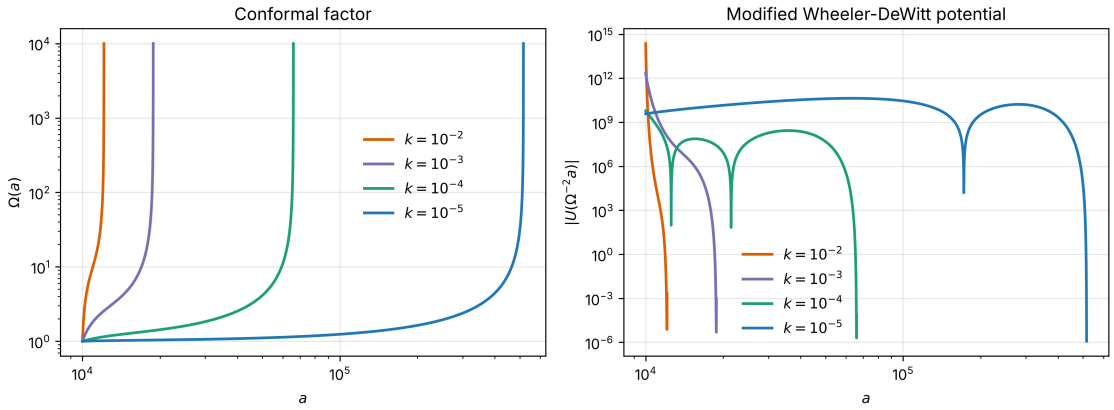


Figure 1. Evolution of the conformal factor Ω (left) and the modified Wheeler-DeWitt potential $|U|$ (right), shown for the initial slopes, $k = d\Omega/da\,(a_i) = 10^{-5}$ (blue), $k = 10^{-4}$ (green), $k = 10^{-3}$ (purple) and $k = 10^{-2}$ (red), with initial conditions $a_i = 10^4$ and $\Omega(a_i) = 1$, in units where $\hbar = c = G = 1$. These solutions describe the region $a \leq a_c$ beyond which Ω remains constant.

perturbations [11, 21–23] within this framework. To do so, we include the perturbation modes ν_k in the Wheeler-DeWitt equation.

$$\left[\frac{1}{24\pi^2}\frac{\partial^2}{\partial a^2} - U(\Omega^{-2}a) - \left[\frac{1}{2}\frac{\partial^2}{\partial \nu_k^2} - \frac{1}{2}\omega_n^2(a)\nu_k^2\right]\right]\Phi = 0. \tag{25}$$

The perturbation modes satisfy the Mukhanov-Sasaki equation,

$$\nu_k'' + \left(k^2 - \frac{\mu^2 - \frac{1}{4}}{\eta^2}\right)\nu_k = 0, \tag{26}$$

where μ is the Hankel index. We wish to study the rescaling of ν under the conformal transformation relating the two successive universes. Since $\mathcal{M}_1$ is DE-dominated at late times, it exhibits a quasi-de Sitter behavior where aH grows. Consequently, modes become super-horizon ($k << aH$) as $\mathcal{M}_1$ expands. The solution to eqn.(26) in the superhorizon limit is given as

$$\nu \sim (-\eta)^{\frac{1}{2}\pm\mu} k^{\pm\mu}. \tag{27}$$

Under conformal transformations, these modes then rescale as,

$$\nu_{\mathbf{k}} \to \Omega^{1\pm 2\mu}\nu_k. \tag{28}$$

Although the Wheeler-DeWitt wave function transforms as $\Phi = J^{-1/2}\Psi$, this overall factor is removed when constructing physical observables. The wave function entering correlation functions is the renormalized perturbation wave function, so the overall normalization of the Wheeler-DeWitt state does not contribute to correlation functions. Consequently, observables such as the primordial power spectrum inherit their conformal transformation solely from the perturbation modes, with no contributions arising from the conformal weight of the Wheeler-DeWitt wave function.

Now, the curvature perturbation is given by

$$\zeta = \sqrt{\frac{4\pi G}{\epsilon}}\frac{\nu_{\mathbf{k}}}{a}, \tag{29}$$

and consequently carries a conformal weight of $3 \pm 2\mu$.

In a deSitter spacetime, we have $\mu = 3/2$ and hence, curvature perturbations have a conformal factor of 0 or +6. Consequently, there exists a branch for which the curvature perturbation is conformally invariant. Since the primordial power spectrum satisfies

$$P_\zeta(k) \propto |\zeta_{\mathbf{k}}|^2, \tag{30}$$

this branch also preserves the power spectrum across the conformal crossover. In other words, the spectral information encoded in the perturbations is transmitted unchanged from one aeon to the next.

Consequently, the initial spectral index

$$n_s - 1 = \frac{d \ln P_\zeta}{d \ln k}, \tag{31}$$

of $\mathcal{M}_2$ is equal that of the final quasi deSitter phase of $\mathcal{M}_1$. For a quasi-de Sitter background defined by slow-roll parameters $\epsilon = -\dot{H}/H^2$, and $\eta = \dot{\epsilon}/H\epsilon$, the standard result is $n_s - 1 = -2\epsilon - \eta$. In the asymptotic deSitter limit ($\epsilon \to 0, \eta \to 0$) we obtain $n_s \to 1$. Hence, $n_s \simeq 1$ follows already in the simplest realization of our framework.

The value $n_s = 0.9649 \pm 0.0042$, measured by Planck [24] indicates a small red tilt in the primordial spectrum of our universe. Within the present framework, such a deviation from unity can be achieved through additional dynamics within $\mathcal{M}_2$ itself or by employing a more general matching functional rather than the preliminary example given in eqn.(7). The central takeaway is that the nearly scale-invariant case emerges already in the simplest example.

By contrast, the opposite branch yields

$$P_\zeta(\mathbf{k}) \to \Omega^{12} P_\zeta(\mathbf{k}). \tag{32}$$

Such a spectrum would correspond to extremely large primordial fluctuations and is therefore physically unacceptable. Consequently, only the conformally invariant branch leads to a well-defined perturbation theory across the conformal crossover.

3 Conclusions

In this work, we have proposed a generalized quantum cosmological framework describing successive universes by extending the gravitational path integral to include geometric correlations between two spacetime manifolds. Rather than treating individual cosmological histories independently, the formalism introduces a matching functional defined on a common crossover surface, allowing information from $\mathcal{M}_1$ to influence the quantum state of its successor. The construction is intentionally general and is not tied to any specific realization of the crossover geometry.

As an illustrative example, we considered the conformal matching prescription of Penrose's Conformal Cyclic Cosmology [17–20], where the metrics of successive universes are related by a conformal transformation. Within this framework, we derived the corresponding minisuperspace Wheeler-DeWitt equation governing the universe of $\mathcal{M}_2$ and obtained the effective Wheeler-DeWitt potential induced by the conformal rescaling. The conformal factor was determined by solving Penrose's Phantom Field Equation.

Our numerical analysis demonstrates that the conformal factor rapidly grows with the scale factor, while the effective Wheeler-DeWitt potential is rapidly suppressed. Consequently, the conformally rescaled wave function evolves into an approximately free Wheeler-DeWitt state as the second universe of $\mathcal{M}_2$ emerges from the crossover. This behavior suggests that, within the conformal realization considered here, the crossover acts as an asymptotically free quantum boundary separating successive cosmological histories.

We also investigated cosmological perturbations [11, 21–23] within the proposed framework. We showed that the physically admissible conformally invariant branch preserves the primordial power spectrum across the crossover surface, whereas the alternative branch produces an extremely large spectrum and is therefore excluded. Because the preceding universe asymptotically approaches de Sitter spacetime, the inherited

perturbations satisfy $n_s \simeq 1$, suggesting that this preliminary model already preserves a scale-invariant primordial spectrum.

The present work should be viewed as a first step toward a more general theory of quantum cosmology involving multiple connected universes. Since the matching functional was introduced in a model-independent manner, choices beyond conformal matching may be explored, potentially leading to different quantum dynamics across the crossover sector of a cyclic picture. Such generalizations, as well as the subsequent evolution of perturbations within $\mathcal{M}_2$ may generate deviations from exact scale invariance $n_s = 1$, as measured by Planck [24].

Acknowledgments

The authors are deeply grateful to Paulo Moniz for his invaluable guidance, insightful discussions, constructive comments, and numerous helpful suggestions, which have significantly contributed to the development of this work. The authors also thank Oem Trivedi for useful discussions and comments.

References

[1] J. A. Wheeler. Geometrodynamics and the issue of final state. In *Les Houches Summer School of Theoretical Physics: Relativity, Groups and Topology*, pages 317–522, 1964.

[2] Bryce S. DeWitt. Quantum theory of gravity. i. the canonical theory. *Phys. Rev.*, 160:1113–1148, Aug 1967.

[3] N. D. Birrell and P. C. W. Davies. *Quantum Fields in Curved Space*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, Cambridge, UK, 1982.

[4] Martin Bojowald. *Quantum Cosmology: A Fundamental Description of the Universe*, volume 835 of *Lecture Notes in Physics*. Springer Science & Business Media, 2011.

[5] J. J. Halliwell. Introductory lectures on quantum cosmology (1990), 2009.

[6] Stephen W. Hawking and George F. R. Ellis. *The Large Scale Structure of Space-Time*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 2 2023.

[7] Shahram Jalalzadeh and Paulo Vargas Moniz. *Challenging Routes in Quantum Cosmology*. World Scientific, Singapore, 2022.

[8] Maxim Yu Khlopov and Oem Trivedi. *Quantum Gravitational Imprints on Cosmology*. World Scientific, 1 2026.

[9] Claus Kiefer. *The semiclassical approximation to quantum gravity*, page 170–212. Springer Berlin Heidelberg.

[10] Claus Kiefer. *Quantum gravity*, volume 124. Clarendon, Oxford, 2004.

[11] Claus Kiefer and David Polarski. Why do cosmological perturbations look classical to us?, 2009.

[12] Paulo Vargas Moniz. *Quantum Cosmology – The Supersymmetric Perspective, Vol. 1: Fundamentals*, volume 803 of *Lecture Notes in Physics*. Springer-Verlag, Berlin, Heidelberg, 2010.

[13] Paulo Vargas Moniz. *Quantum Cosmology – The Supersymmetric Perspective, Vol. 2: Advanced Topics*, volume 804 of *Lecture Notes in Physics*. Springer-Verlag, Berlin Heidelberg, 2010. eBook ISBN: 978-3-642-11570-7.

[14] J. B. Hartle and S. W. Hawking. Wave function of the universe. *Phys. Rev. D*, 28:2960–2975, Dec 1983.

[15] Alexander Vilenkin. Quantum creation of universes. *Phys. Rev. D*, 30:509(R)–511(R), Jul 1984.

[16] Aarav Shah, Paulo Moniz, Oem Trivedi, and Meet J. Vyas. Non-markovian memory-induced effects in quantum cosmology, 2026.

[17] Roger Penrose. *Cycles of Time: An Extraordinary New View of the Universe*. Vintage Books, New York, NY, 2011.

[18] Roger Penrose. On the Gravitization of Quantum Mechanics 2: Conformal Cyclic Cosmology. *Found. Phys.*, 44(8):873–890, 2014.

[19] Krzysztof A. Meissner and Roger Penrose. The physics of conformal cyclic cosmology, 2025.

[20] Oliver Markwell and Chris Stevens. Toward fixing a framework for conformal cyclic cosmology. *General Relativity and Gravitation*, 55:1–14, 2022.

[21] V. F. Mukhanov, H. A. Feldman, and Robert H. Brandenberger. Theory of cosmological perturbations. *Phys. Rept.*, 215:203–333, 1992.

[22] V. Mukhanov. *Physical Foundations of Cosmology*. Cambridge University Press, Oxford, 2005.

[23] J. J. Halliwell and S. W. Hawking. The Origin of Structure in the Universe. *Phys. Rev. D*, 31:1777, 1985.

[24] Planck Collaboration. Planck 2018 results: Vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.

[25] Paul Tod. The equations of Conformal Cyclic Cosmology. *General Relativity and Gravitation*, 47:17, March 2015.

[26] Ezra T. Newman. A fundamental solution to the ccc equations. *General Relativity and Gravitation*, 46(5), April 2014.

[27] Paweł Nurowski. Poincare–Einstein approach to Penrose's conformal cyclic cosmology. *Class. Quant. Grav.*, 38(14):145004, 2021.